Nahyun Lee

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Boston, MA, United States

Research Interests

Augmenting how we perceive the world.

My research investigates human perception and sensemaking, drawing on **HCI, XR/haptics, and behavioral research.**

I aim to understand the cognitive and neural mechanisms underlying perception and decision-making, and translate these insights into systems that augment human cognition.

Education

University of Massachusetts Boston *Sep 2026 – Jun 2027*
Exchange Student in Computer Science Boston, MA, USA

Sungkyunkwan University *Feb 2023 – Expected Feb 2028*
B.S. in Mechanical Engineering, Software Engineering, and Human-Computer Interaction Suwon, South Korea
◦ GPA: 4.35/4.50

Hanyang University *Mar 2022 – Dec 2022*
Undergraduate Coursework in Business Administration Seoul, South Korea

Research Experience

Wearable and Interactive Technology Lab, KAIST *Jan 2026 – Sep 2026*
Research Intern | *Advised by* ***Prof. Ian Oakley*** | *Collaborated with* ***Prof. Sangho Yoon*** *(HCI Tech Lab, KAIST)* Daejeon, South Korea
◦ Developed movement-coupled vibrotactile feedback using granular synthesis, exploring audiovisual-tactile illusions.
◦ Designed and conducted 2AFC and VR interaction studies, identifying task-dependent effects on haptic experience.

Future Computing Lab, Sungkyunkwan University *Sep 2025 – Sep 2026*
Research Intern | *Advised by* ***Prof. Sunjae Lee*** | *Collaborated with* ***Dr. Hyeon Jeon*** *(Seoul National University)* Suwon, South Korea
◦ Developed interactive systems to visualize missing evidence in investment decisions and support reflection on losses.
◦ Derived a five-stage loss-acceptance framework through formative interviews and evaluated a reflection system.

Independent Research, Stanford Center at the Incheon Global Campus *Mar 2024 – Dec 2025*
Research Assistant | *Advised by* ***Hoseok Jung*** | *Collaborated with Imperial College London and TU Wien* Incheon, South Korea
◦ Developed wearable and encountered-type thermal haptic interfaces for VR firefighting and Mars exploration training.
◦ Built a Fire Dynamics Simulator-Unity pipeline to translate fire conditions into spatially varying thermal feedback.

Publications

J=Journal, C=Conference, P=Poster, U=Under Review

[U.4] **Nahyun Lee***, Sunjae Lee, and Hyeon Jeon.
FacingLoss: Facilitating Learning from Stock Investment Losses through Structured Reflection

[U.3] Sara Scheffer*, Elias Panner, **Nahyun Lee**, Hoseok Jung, Wanting Mao, Sebastian Lux, Frank Long, Arnab Majumdar, and Kurt Matyas.
Feeling the Heat: Embodied Multimodal Mixed Reality for Firefighter Sensemaking and Cognitive Load Management

[U.2] Min Htet Myint*, **Nahyun Lee**, Sungmin Yoon.
A VR-based digital twin framework for spatially grounded BIM-MEP HVAC operation and maintenance with role-separated LLM-agent services

[U.1] Sara Scheffer*, Elias Panner, **Nahyun Lee**, Hoseok Jung, Wanting Mao, Sebastian Lux, Frank Long, Arnab Majumdar, and Kurt Matyas.
Embodied Multimodal Mixed Reality for Firefighter Training: Evaluating Visual-Thermal Integration for Situational Awareness and Decision Performance

[P.2] Hoseok Jung*, Jiyoon Lee, **Nahyun Lee**, and Hyunmin Kang.
Fire Dynamics Data-Driven Thermal Haptics for VR Fire Simulation
The 32nd ACM Symposium on Virtual Reality Software and Technology (VRST 2026 Posters)

[C.6] Hoseok Jung*, Jiyoon Lee, Sungmo Lee, **Nahyun Lee**, Seungwon Hu, Boyoung Kim, Okkeun Lee, Hyochang Kim, and Hyunmin Kang (2026).
A Wearable Encountered-Type ThermalKinesthetic Interface for Mars Exploration VR
28th International Conference on Human-Computer Interaction (HCII 2026 Invited Papers)

[P.1] **Nahyun Lee*** and Hyeon Jeon (2026).
Is This Enough to Decide? Facilitating Informed Investment Decisions by Visualizing Missing Evidence
ACM Conference on Human Factors in Computing Systems Extended Abstracts (CHI 2026 Extended Abstract)

[C.5] Myeongjun Kim*, Seungmin Lee*, Junjae Park*, **Nahyun Lee***, and Jo Woon Chong (2025).
A Low-Cost Quantitative Evaluation Testbed of Small-Sized IMU Sensor Modules Using a Smartphone
10th International Conference on Consumer Electronics Asia (ICCE-Asia 2025)

[C.4] Hoseok Jung*, Jiyoon Lee, Sungmo Lee, and **Nahyun Lee** (2025).
Encounter-type Force and Thermal Haptics for VR with a Wrist-Worn Articulated Robot
Korea Society of Design Science, 2025 Fall International Conference (KSDS 2025)

[C.3] Hoseok Jung*, Dongho Shin*, **Nahyun Lee***, Minseo Yu*, Kyuhyoung Cho*, and Gyungsun Go* (2024).
Mars Exploration Simulation with an Encounter-type Multimodal Haptic Device
Korea Society of Design Science, 2024 Spring International Conference (KSDS 2024)

[C.2] Jiyoon Lee*, Jiwon Yang, ChungHyun Lee, **Nahyun Lee**, Hyunmin Kang, and Hoseok Jung (2024).
Proposal for a Virtual Reality Mars Exploration Exhibit Utilizing Reverse-Temperature Multi-Modal Haptic Devices for Popularizing Space Science
International Symposium of Science Museums 2024 (ISSM 2024)

[C.1] Doohwan Kim*, Yohan Han, Inhyuk Jeong, Yeongseok Hwang, Jinju Park, **Nahyun Lee**, and Yujin Lee (2024).
LLM-based chatbot system to improve worker efficiency and prevent safety incidents
Korea Society of Computer Information, 2024 Winter Conference (KSCI 2024)

Academic Service

Reviewer

◦ ACM International Joint Conference on Pervasive and Ubiquitous Computing (Ubicomp 2026)

◦ ACM Conference on Human Factors in Computing Systems (CHI 2026)

Student Volunteers

◦ ACM Symposium on User Interface Software and Technology (UIST 2026)

◦ IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2025)

Skills

Research Methods: User Studies, Behavioral Experiments, Quantitative and Qualitative Analysis

Software & AI Development: Python, C/C++, MATLAB, React, Next.js, Flutter, LLM-based applications

XR & Hardware Prototyping: Unity (VR/AR), Haptic and Wearable Systems, Sensor Integration, Arduino, 3D printing

Startup Experience

Modly *Feb 2025 – Jan 2026*
Co-Founder Suwon, South Korea

◦ Selected for the Korean I-Corps and completed technology commercialization training at George Washington University.

◦ Led business strategy for a 3D scanning service, conducting 100+ user interviews to identify viable applications.

AiGENDA [🌐] *Jun 2024 – Aug 2024*
AI Developer Suwon, South Korea

◦ Developed a lecture-based video Q&A system using speech-to-text (STT), LLMs, and LangChain.

◦ Improved domain-specific term recognition accuracy (60% → 98%) through retrieval-augmented transcript correction.

Teaching & Leadership Experience

Data Structures Tutor – International Student Tutoring Program *Sep 2025 – Dec 2025*
Sungkyunkwan University Suwon, South Korea

XR Systems & AI Integration Seminar Lead *Jul 2025 – Sep 2025*
Building Information Science & Technology Lab (BIST), Sungkyunkwan University Suwon, South Korea
◦ Led lab seminars on XR development (Unity, Meta Quest) and AI integration for architectural engineering applications.

President – KingoSmartThings *Dec 2024 – Feb 2026*
AI/IoT Entrepreneurship Student Organization, Sungkyunkwan University Suwon, South Korea
◦ Led the organization to an Excellence Award in SKKU's Excellent Study Club Program.
◦ Organized campus-wide hackathons (30+ participants) in collaboration with SKKU RISE Center and ICT College.

Honors

Associate Member – Sigma Xi, The Scientific Research Honor Society *Sep 2026*
Sigma Xi

President's List *Sep 2026*
Sungkyunkwan University

University Representative – Young Engineers Honor Society (YEHS) *Dec 2024*
National Academy of Engineering of Korea

Awards

Gold Collar Engineering Student Award *Sep 2026*
Korean Society for Engineering Education

President's Award – URP III (Undergraduate Interdisciplinary Research Program) *Dec 2025*
Institute for Cross-disciplinary Studies, Sungkyunkwan University

Korean I-Corps – Selected for U.S. Entrepreneurship Training *Jun 2025*
Ministry of Science and ICT & National Research Foundation of Korea

Excellence Award – Embedded Software Competition (Open Innovation Track) *Nov 2024*
Ministry of Trade, Industry and Energy & Korea Electronics Technology Institute

Engineering Innovation Award – Samsung Medical Service Design Program *Jan 2024*
Bio-Digital Healthcare Consortium, Sungkyunkwan University

Scholarships

Korea-U.S. Advanced Technology Youth Exchange Support Program (~$18,000) *Jun 2026 – Present*
Korea Institute for Advancement of Technology

Presidential Science Scholarship (~$18,500) *Jul 2025 – Present*
Ministry of Science and ICT

Cho Byung-Doo Scholarship (~$5,400) *Apr 2025 – Sep 2025*
Cho Byung-Doo Scholarship Foundation

Korea Entegris Scholarship (~$2,300) *Sep 2024*
Entegris

Hyeong-Ae Scholarship (~$4,600) *Mar 2024 – Sep 2024*
Hyeong-Ae Scholarship Foundation

Top 3% Academic Excellence Scholarship (~$4,200) *Sep 2023 – Feb 2024*
Sungkyunkwan University

Student Success Creativity Scholarship (~$3,500) *May 2023*
Sungkyunkwan University

Additional Information

Languages: Korean (Native) and English (Full professional)
Interests: Birdwatching and Financial Planning